



Basic Principles of Medicine 1
Module: Musculoskeletal System | Lecture 06

Clinical Anatomy of the Wrist & Hand

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Recommended Reading

- Drake, Vogl, Mitchell. Grays Anatomy for Students. 4th ed. Elsevier Churchill Livingstone.
- Printed Textbook: Chapter 7: pg 753 - 809
- Online Textbook: Chapter 7: [click here](#)
 - Conceptual Overview
 - Regional Anatomy:
 - Hand
 - Motor functions of median and ulnar nerves
- Pay special attention to the **green (printed text)** / **blue (ebook)** “*In the Clinic*” boxes for the clinical correlations for this topic

Objectives

SOM.MK.II.BPM1.1.MSK.1.ANAT.1174

Describe the movements that occur at the wrist joint

SOM.MK.II.BPM1.1.MSK.1.ANAT.1175

Describe the bony and ligamentous anatomy of the wrist joint.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1176

Describe the carpal tunnel and Guyon's canal

SOM.MK.II.BPM1.1.MSK.1.ANAT.1177

Describe the articulations between the distal radius and the proximal row of carpal bones.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1178

Describe the general arrangement between the distal row of carpal bones and the metacarpals (4 distal carpals, 5 metacarpals).

SOM.MK.II.BPM1.1.MSK.1.ANAT.1179

List the most commonly fractured carpal bone

SOM.MK.II.BPM1.1.MSK.1.ANAT.1180

List the most commonly dislocated bone in the carpus and how it can lead to carpal tunnel syndrome

SOM.MK.II.BPM1.1.MSK.1.ANAT.1183

Describe the fascia of the wrist and hand including the flexor and extensor retinaculum, palmar carpal ligament and palmar aponeurosis

SOM.MK.II.BPM1.1.MSK.1.ANAT.1184

Describe the fibrous digital tendon sheaths and their function in preventing bowing of tendons

SOM.MK.II.BPM1.1.MSK.1.ANAT.1185

Describe the anatomical and clinical mechanism of De Quervains syndrome (Grey's Anatomy "In the Clinic" box).

SOM.MK.II.BPM1.1.MSK.1.ANAT.1186

Describe the formation and function of the extensor hoods

SOM.MK.II.BPM1.1.MSK.1.ANAT.1187

Describe the synovial sheaths in the hand and fingers and their importance in the spread of infections.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1188

Describe the presentation of Dupuytren's contracture and "trigger finger" (stenosing tenosynovitis).

SOM.MK.II.BPM1.1.MSK.1.ANAT.1189

List the attachments, innervations and actions of the intrinsic muscles of the hand.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1190

Describe the movements occurring at the carpometacarpal, metacarpophalangeal and interphalangeal joints of the thumb and of digits 2-5.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1191

Discuss the anatomical basis for Allen's test, paying specific attention to the origin of deep and superficial palmar arches.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1192

Describe the boundaries and contents of the "anatomical snuff box".

SOM.MK.II.BPM1.1.MSK.1.ANAT.1193

Describe the cutaneous innervation of the hand

SOM.MK.II.BPM1.1.MSK.1.ANAT.1194

Describe the dermatomes of the upper limb

SOM.MK.II.BPM1.1.MSK.1.ANAT.1197

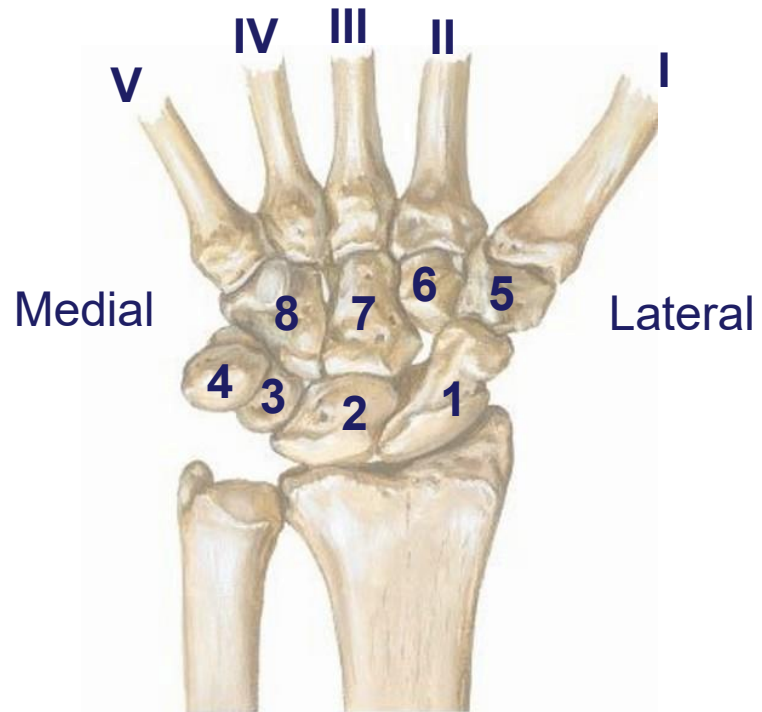
Describe the venous and lymphatic drainage of the arm, forearm and hand.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1198

List the sites commonly used for venipuncture in the upper limb

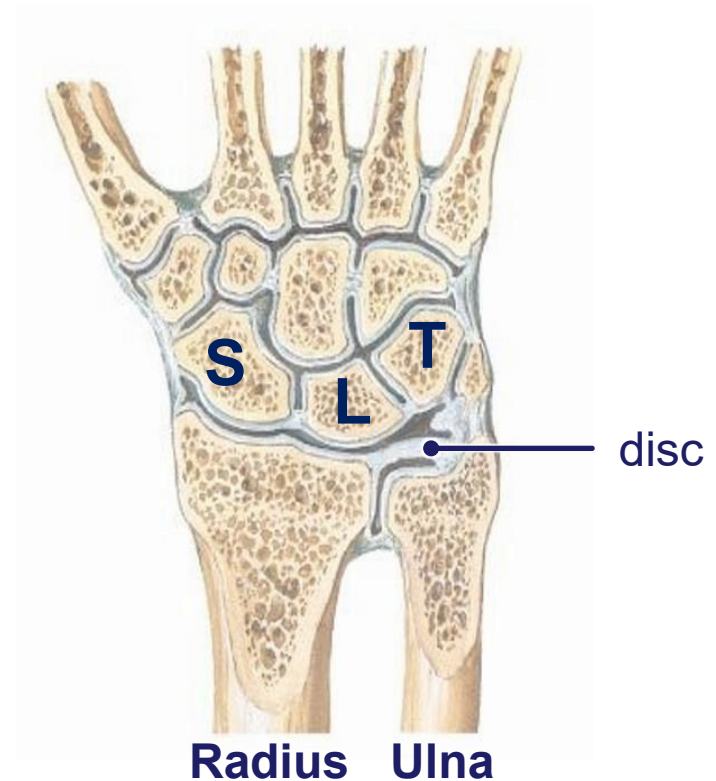
Bones of the Wrist

- Proximal Row – lateral to medial
 - 1) Scaphoid
 - 2) Lunate
 - 3) Triquetrum
 - 4) Pisiform
- Distal row – lateral to medial
 - 5) Trapezium
 - 6) Trapezoid
 - 7) Capitate
 - Articulates with base metacarpal III
 - 8) Hamate
 - Hook of palmar surface



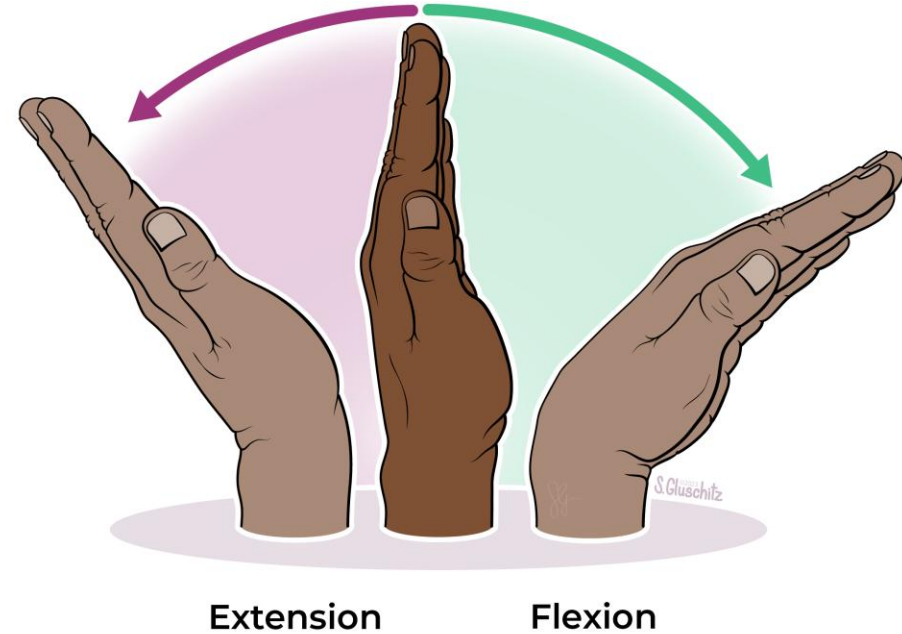
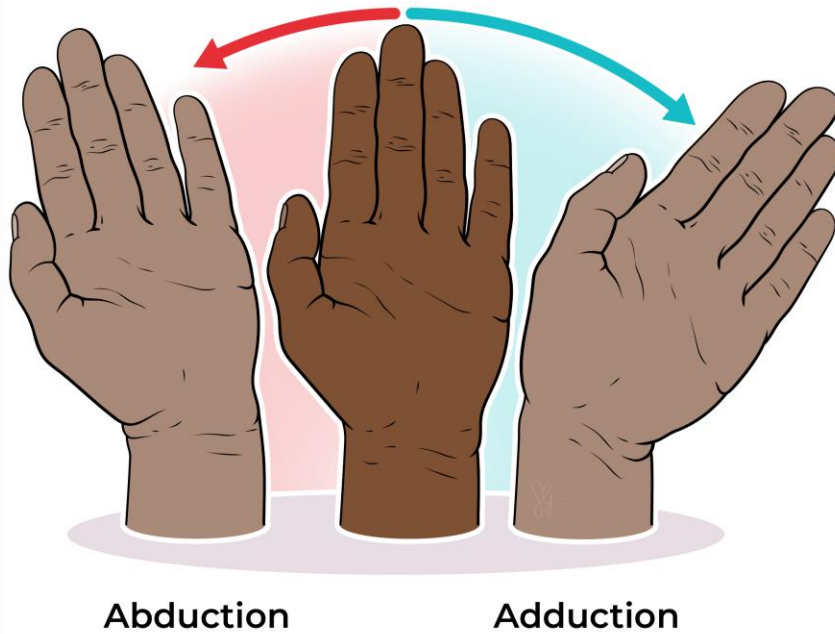
Wrist Joint

- Synovial joint between the distal end of the radius and ulnar with the carpal bones
 - Radius articulates with ***lunate and scaphoid***
 - Radioulnar disc articulates with ***lunate and triquetrum***
- Allows movement of the hand around two axes
 - Flexion and extension
 - Abduction (lateral deviation) and adduction (medial deviation)
- Intercarpal joints
 - Synovial joints share a single capsule
 - Limited movement, but contribute to the positioning of the hand in **flexion, extension, abduction**



Movements at the Wrist Joint

- aka moving the hand



Carpal Tunnel

- The carpal tunnel is formed anteriorly at the wrist

- Floor

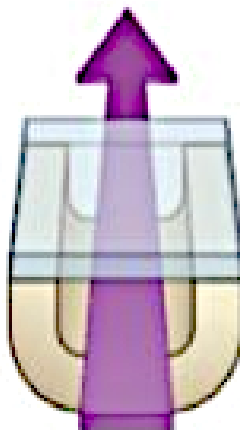
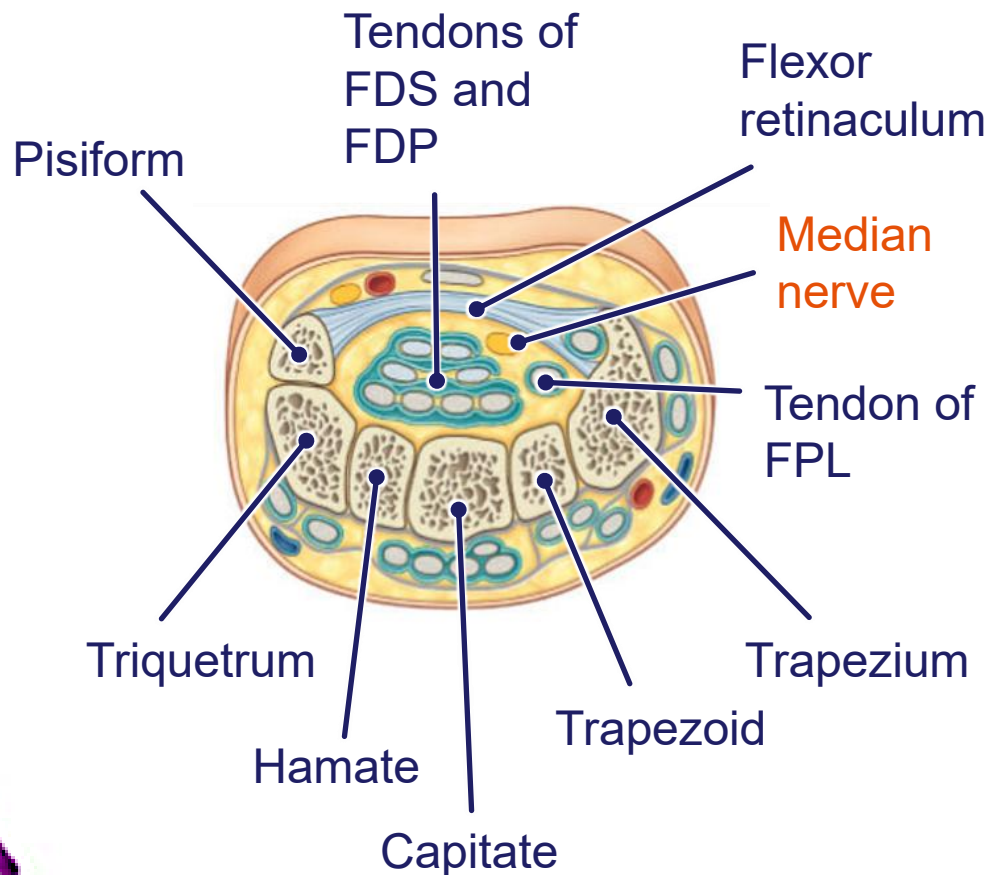
- Deep arch formed by carpal bones

- Roof

- Flexor retinaculum

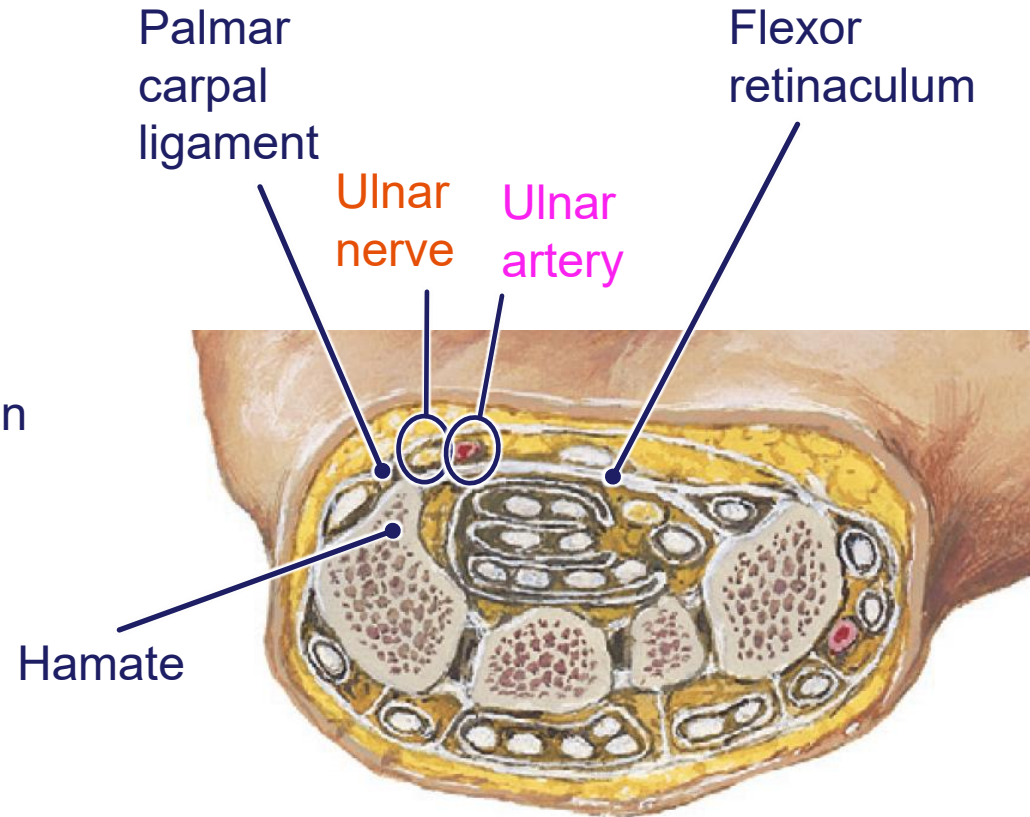
- Contents

- Tendons of FDS, FDP, FPL
 - Median nerve

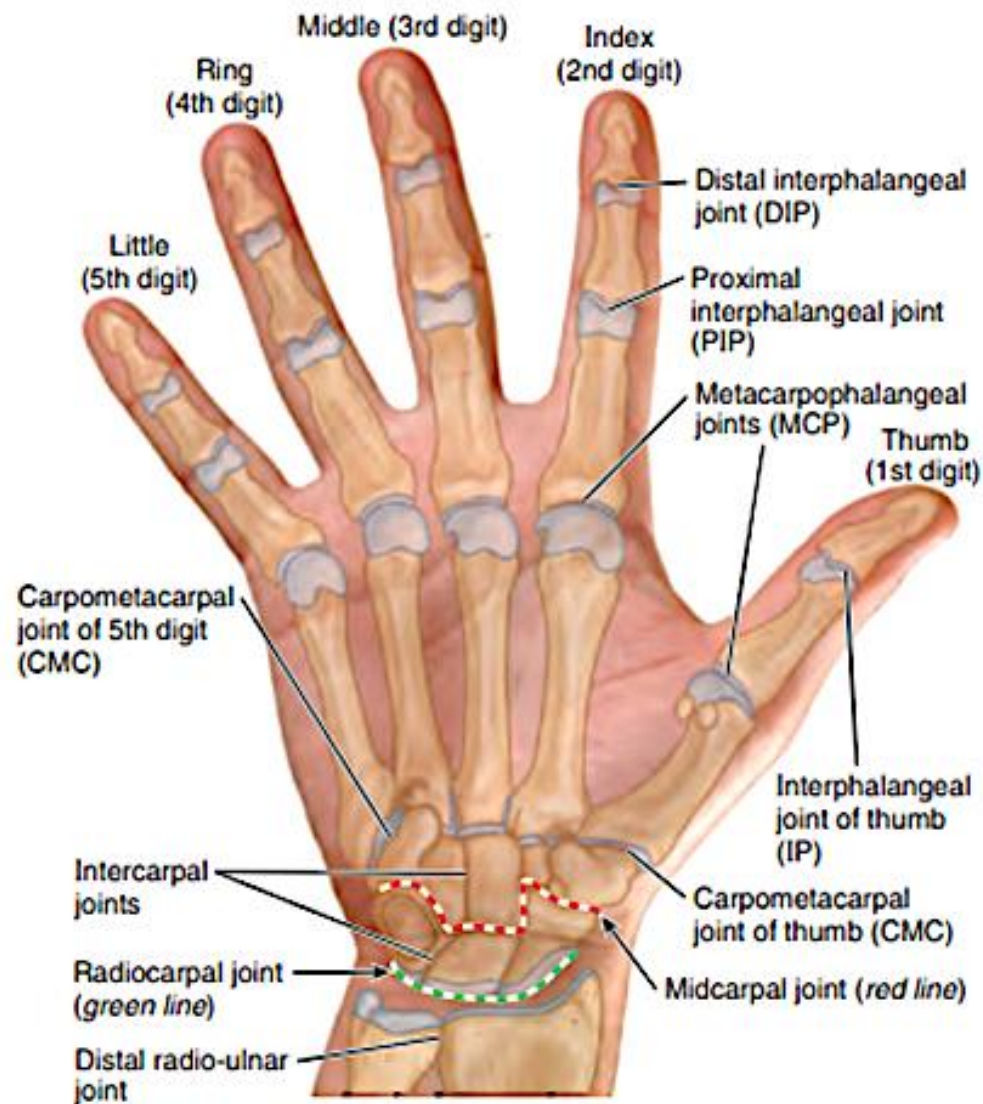


Guyon's Canal

- Roof
 - Palmar carpal ligament
- Floor
 - Flexor retinaculum
 - Hypothenar muscles
- Contents
 - Ulnar nerve, artery, and vein



Joints and Bones of the Hand



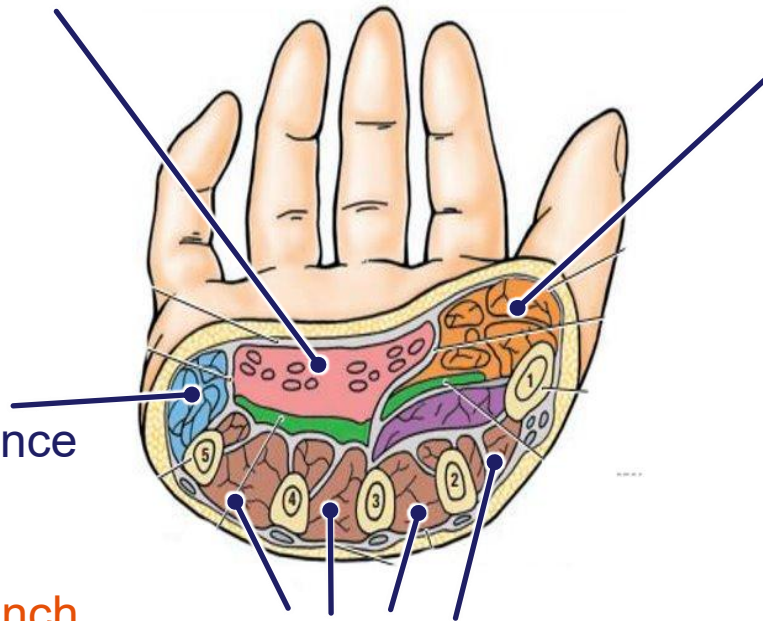
Compartments of the Hand

Midpalmar (central)

- Contains long flexor tendons, lumbricals, and **palmar arches**

Hypothenar eminence

- Contains 3 little finger muscles
- **Deep motor branch of ulnar nerve**



Thenar eminence

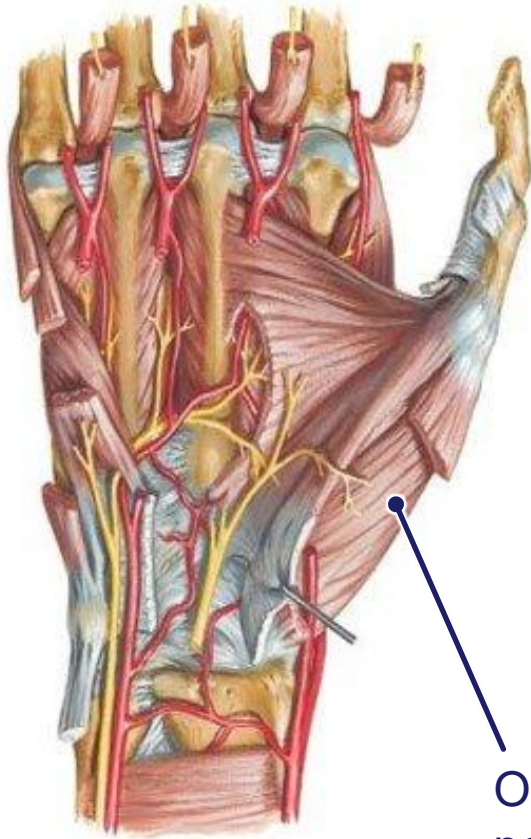
- Contains FPL tendon and 3 thumb muscles
- **Recurrent median nerve**

Four interosseous spaces

- Palmar and dorsal IO muscles
- **Deep motor division of the ulnar nerve**

Intrinsic Muscles of the Hand: Thenar

Innervated by **recurrent**
branch of the median nerve



Opponens
pollicis



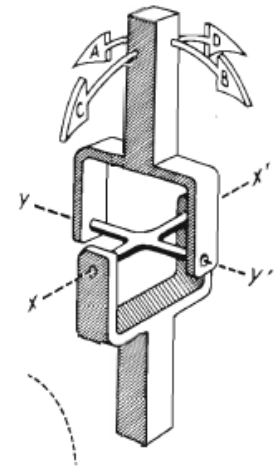
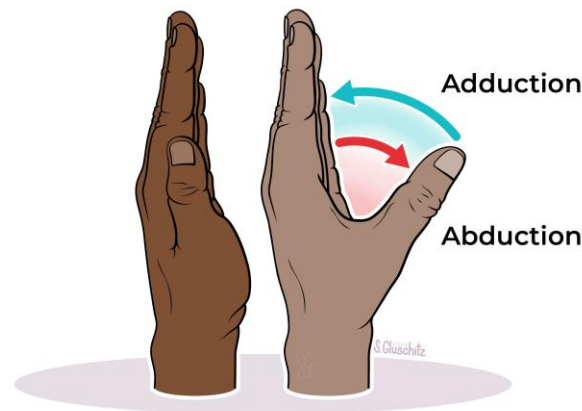
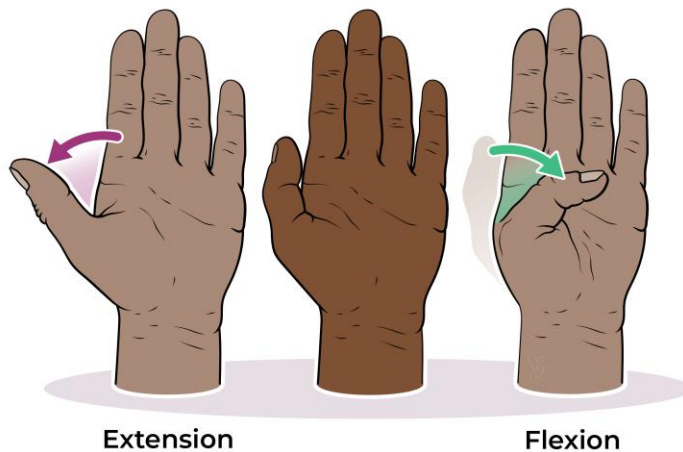
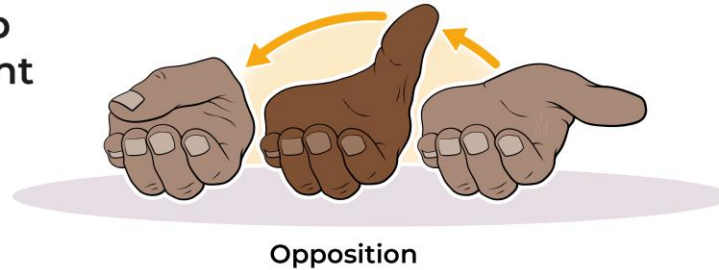
Flexor
pollicis
brevis

Abductor
pollicis
brevis

Movements of the Thumb

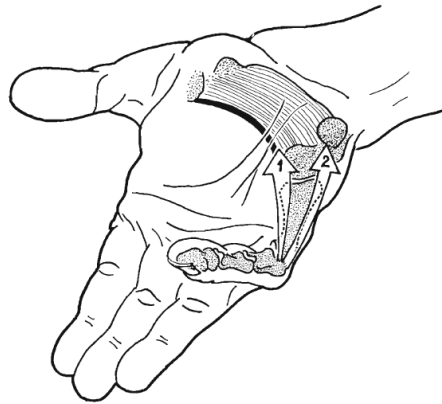
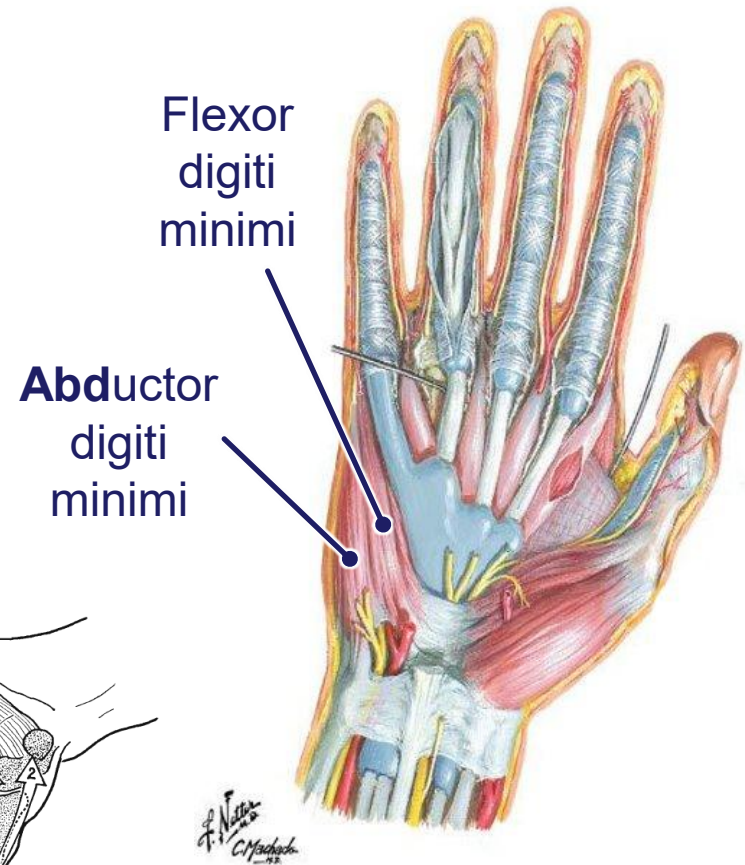
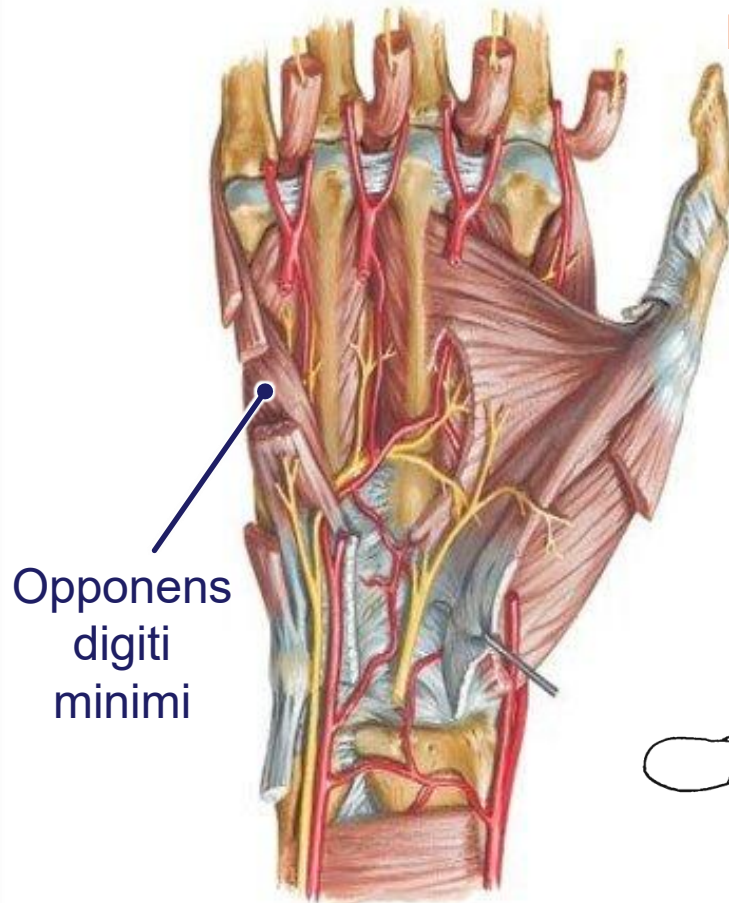
- 1st carpometacarpal (CMC) joint between trapezium and base of thumb
- Saddle joint that permits **flexion**, **extension**, **abduction**, **adduction** and **opposition** of the thumb

Movements of the thumb are at 90° to the movement of the digits



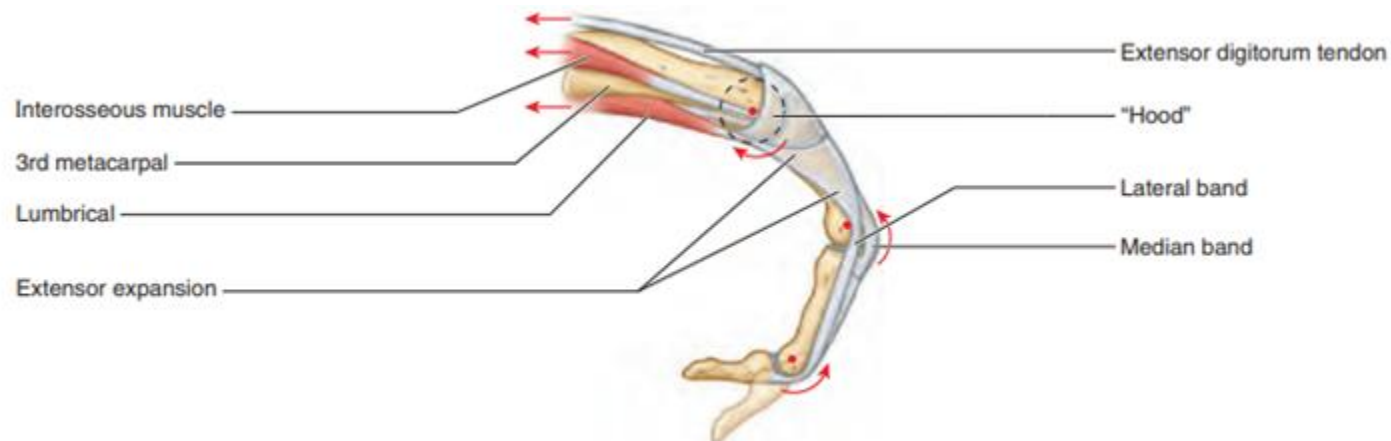
Intrinsic Muscles of the Hand: Hypothenar

Innervated by **deep**
branch of the ulnar nerve

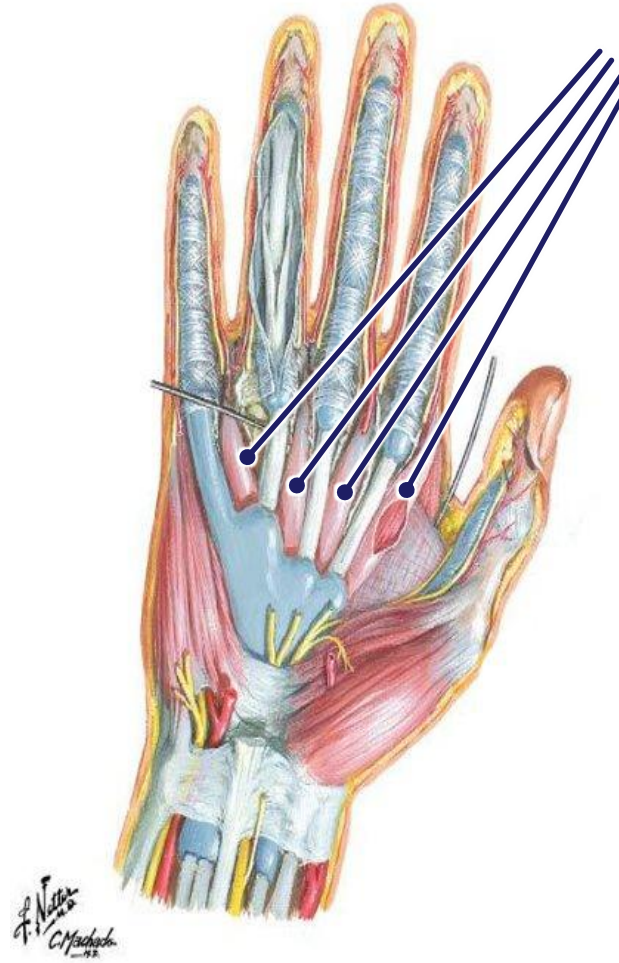
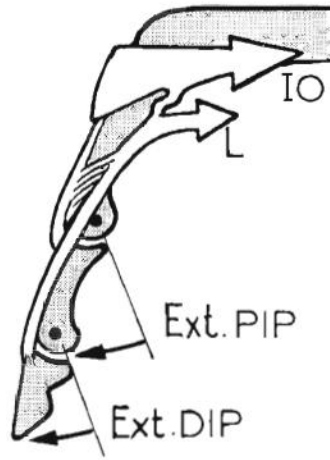


Extensor Hood

- Tendons of **extensor digitorum** and **extensor pollicis longus** expand over the proximal phalanx
 - Makes a “hood” that then divides into a central band and two lateral bands
- Serves as an attachment for:
 - **Lumbricals**
 - **Dorsal interossei**
 - **Palmar interossei**
- Due to attachments, these muscles also **extend** at the interphalangeal joint

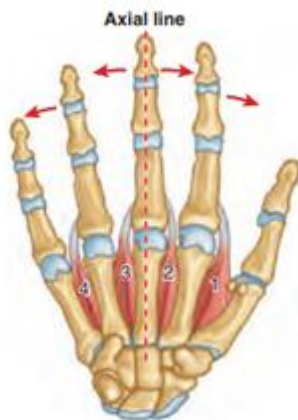


Intrinsic Muscles of the Hand: Lumbricals



- 1st - 2nd: palmar digital branches of **median nerve**
- 3rd - 4th: **deep branch of ulnar nerve**
- **Flex MCP joints and extend IP joints of digits 2-5**

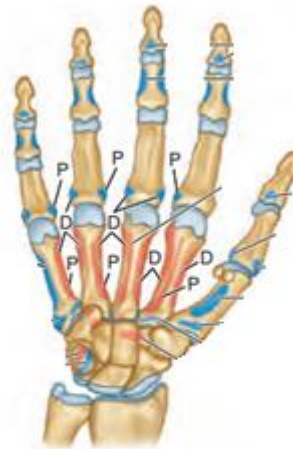
Intrinsic Muscles of the Hand: Interossei



(D) Dorsal interossei (1-4)
Palmar views

Dorsal interosseus

- **Abduction** of digits
- **Extension** of digits
- **Deep branch of the ulnar nerve**



(A) Palmar view

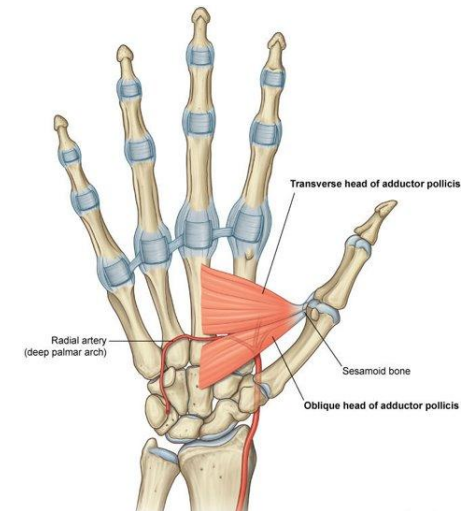
P = Palmar interossei
D = Dorsal interossei



(E) Palmar interossei (1-3)
Palmar views

Palmar interosseus

- **Adduction** of digits
- **Extension** of digits
- **Deep branch of the ulnar nerve**



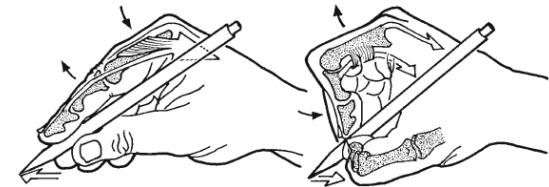
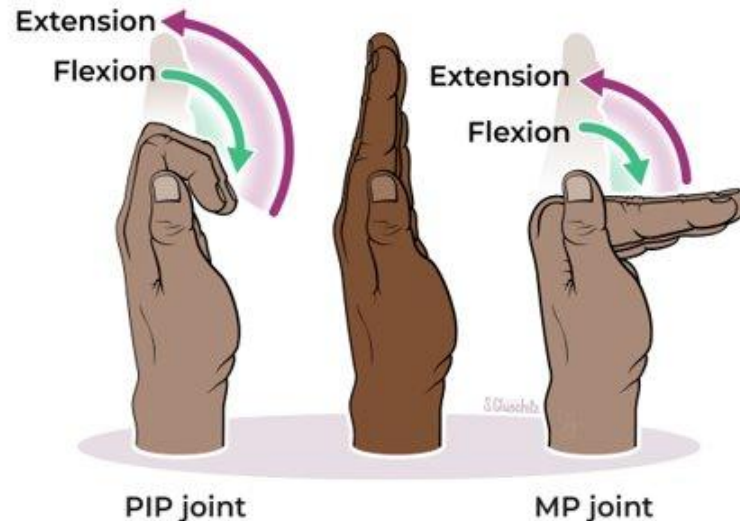
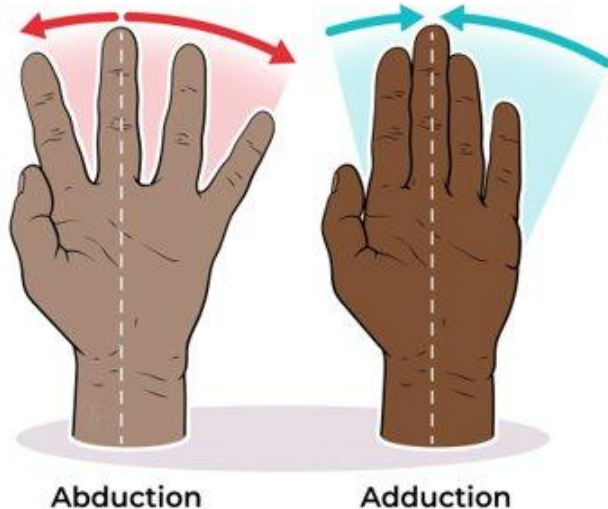
Adductor pollicis

- **Adducts the thumb**
- **Deep branch of the ulnar nerve**

Movements of the digits

- 2nd to 5th carpometacarpal (CMC) joints only permit limited **gliding** movements
- Metacarpophalangeal (MCP) joints permit **flexion**, **extension**, **abduction**, and **adduction** of digits
- Interphalangeal (IP) joints permit **flexion and extension**

Abduction and adduction
occur around digit III



Tendon Sheaths

Fibrous digital tendon sheaths

- Dense CT surrounding tendons of **FDP**, **FDS** and **FPL** muscles
- Hold tendons to phalanges and **prevent bowing**

Synovial sheaths

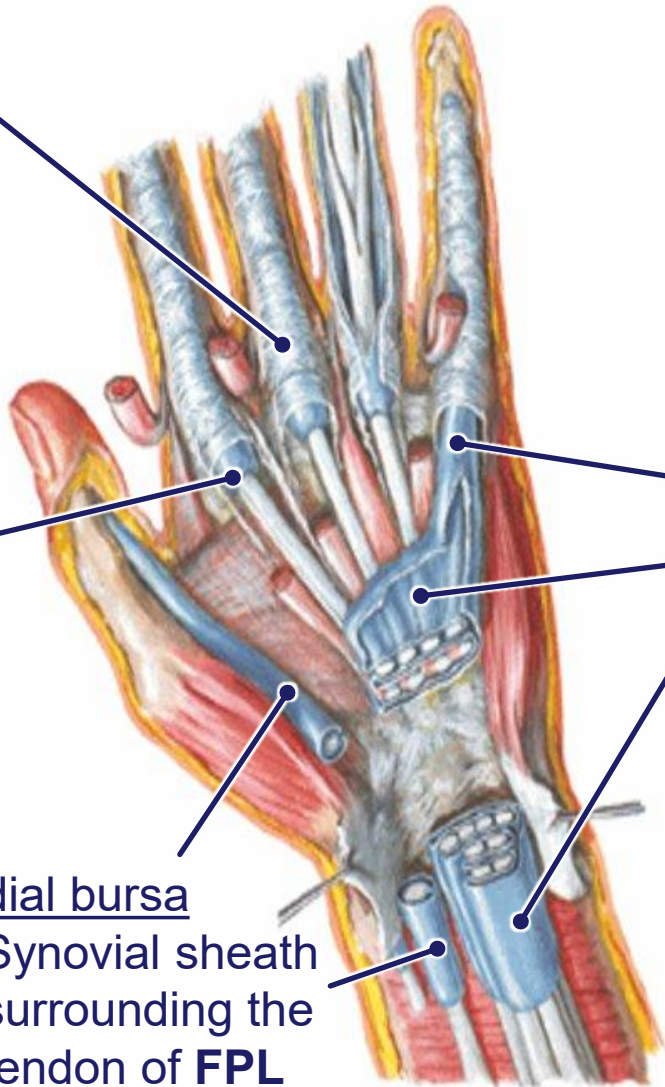
- Within the fibrous sheath the tendon has a synovial sheath directly surrounding it

Radial bursa

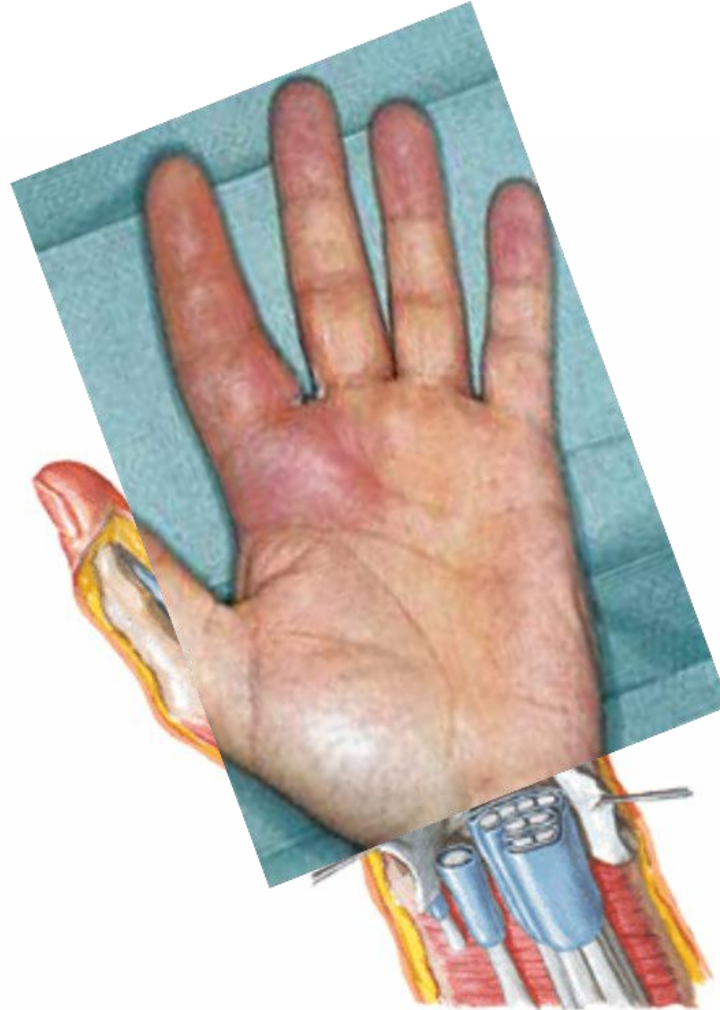
- Synovial sheath surrounding the tendon of **FPL** continuing through the **carpal tunnel**

Ulnar bursa – common flexor sheath

- Synovial sheath surrounding the tendons of **FDP** and **FDS of 5th digit** continue through the midpalmar space and into the **carpal tunnel**



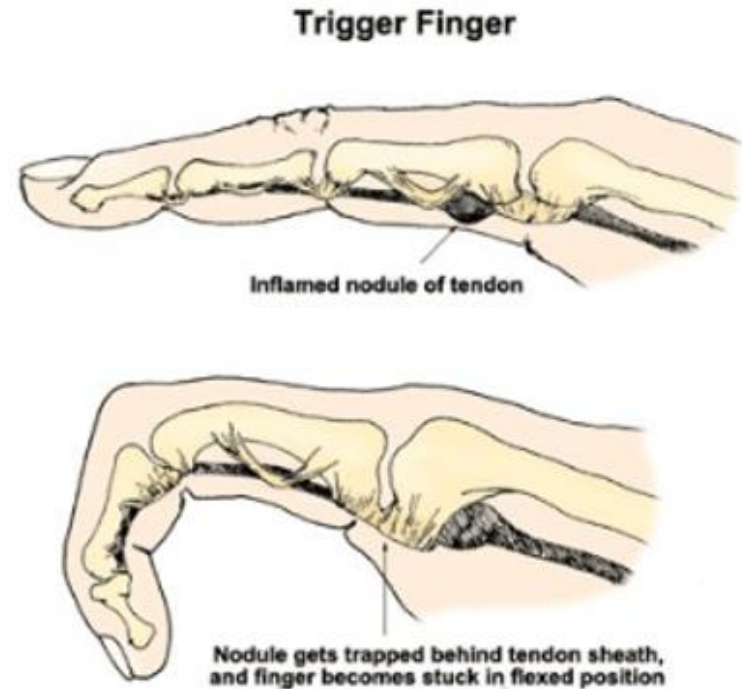
Clinical Correlate: Inflammation of Synovial Sheaths



Clinical Correlate:

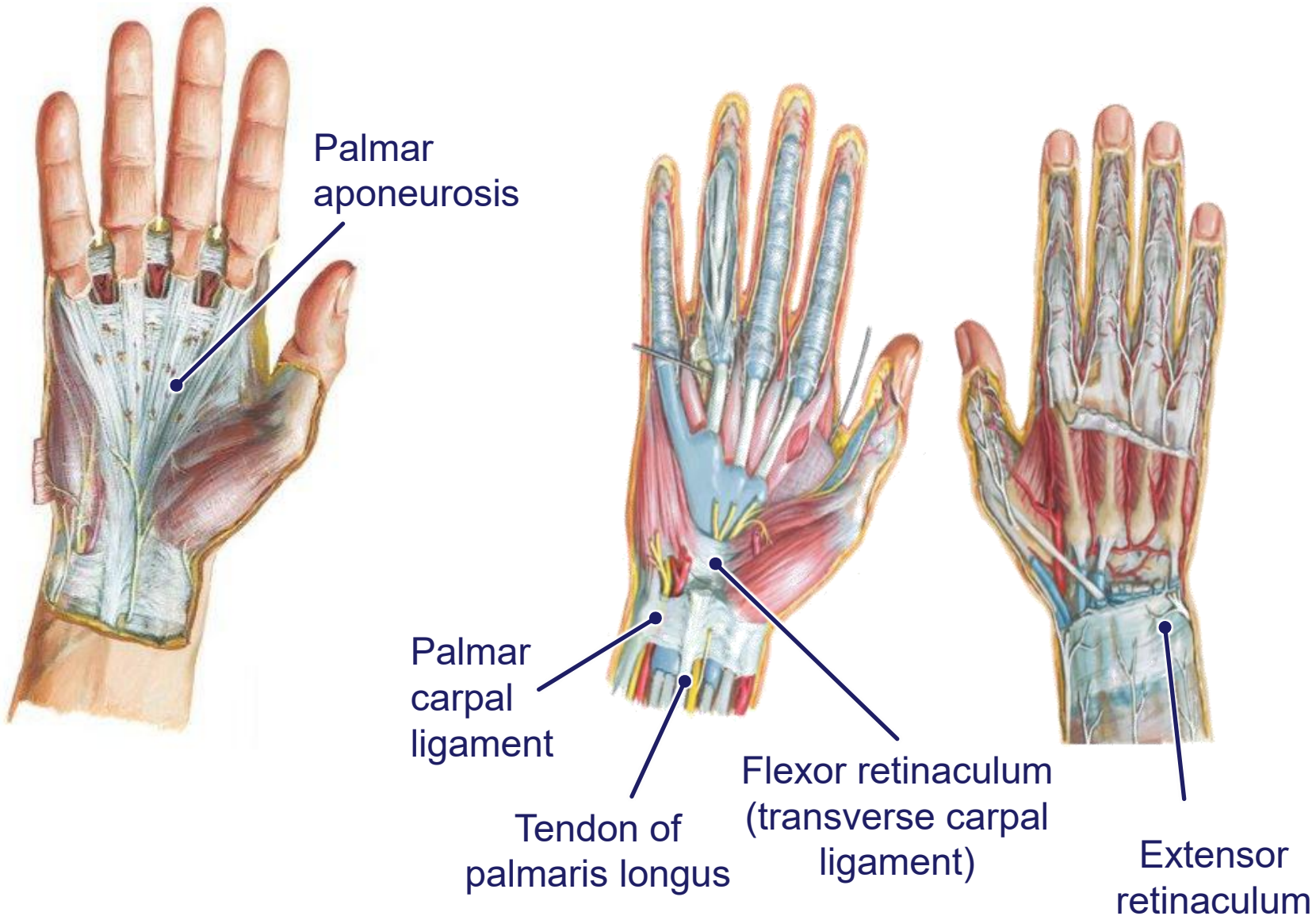
Trigger Finger (Stenosing Tenosynovitis)

- Fibrosis and tightening of the fibrous digital sheath of the flexor tendons at the level of the **MCP joint**
- Inflammation and nodule formation of the flexor tendons
- Triggering of the affected finger is due to tendons requiring **excessive force to fully extend or flex**



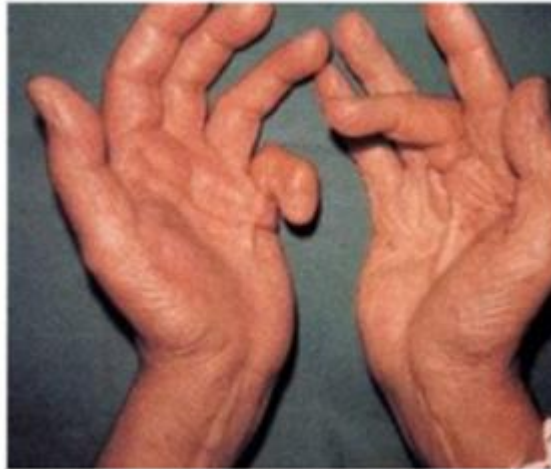
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Fascia of the Hand



Clinical Correlate: Dupuytren's contracture

- **Flexion contracture** of the hand
- Thickening and contraction of the palmar aponeurosis
- Nodule formation
- **Unable to extend fingers**



Needle aponeurotomy: minimally invasive technique where the cords are weakened through the insertion and manipulation of a small needle. Once weakened, the offending cords may be snapped by simply pulling the finger(s) straight

Invasive surgery: cutting out thickened and contracted bands in palmar aponeurosis

Anatomical Snuff Box

- A triangular depression formed on the posterolateral side of the wrist
- Lateral border
 - Abductor pollicis longus
 - Extensor pollicis brevis
- Medial border
 - Extensor pollicis longus
- Roof
 - Superficial branch of the radial nerve
 - Cephalic vein
- Floor
 - Scaphoid and trapezium
 - Distal ends of the tendons of ECRL and ECRB
- Contents
 - Radial artery

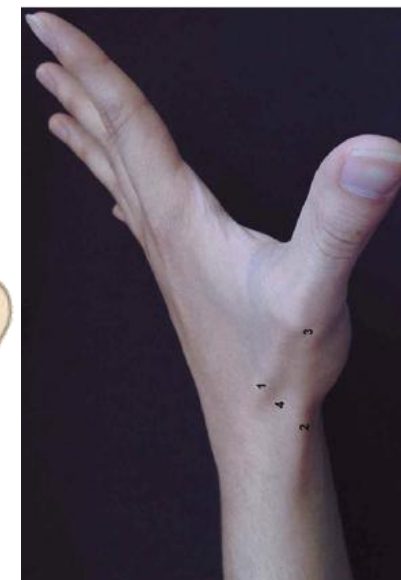
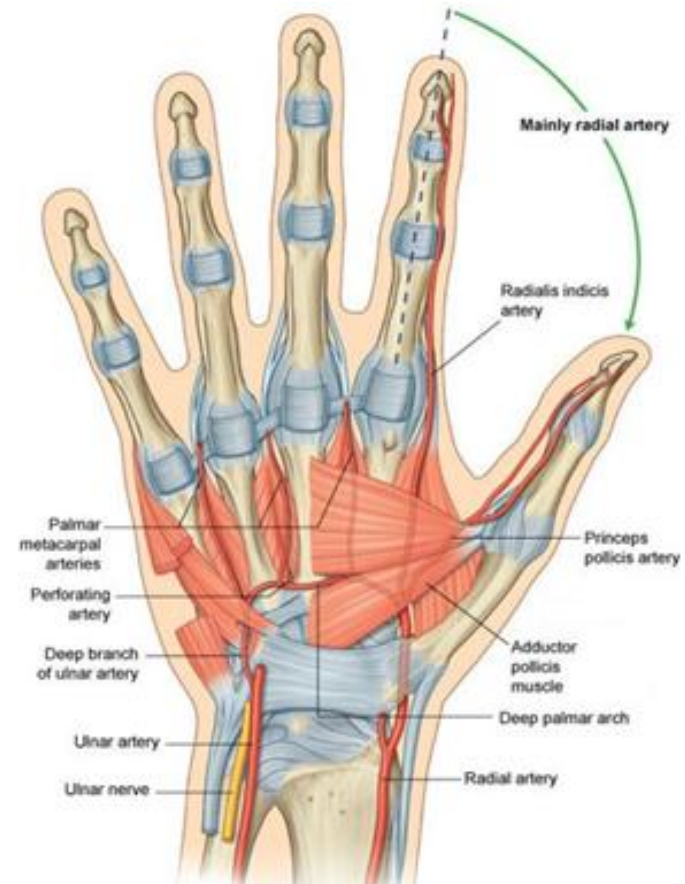
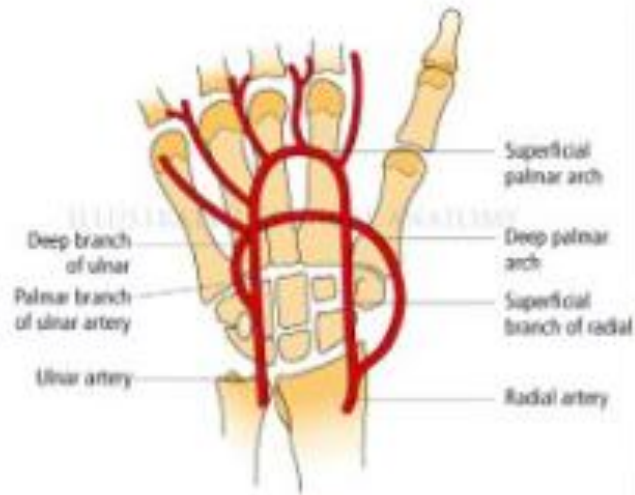


Fig. 1

Blood Supply



Clinical Correlate: Allen Test



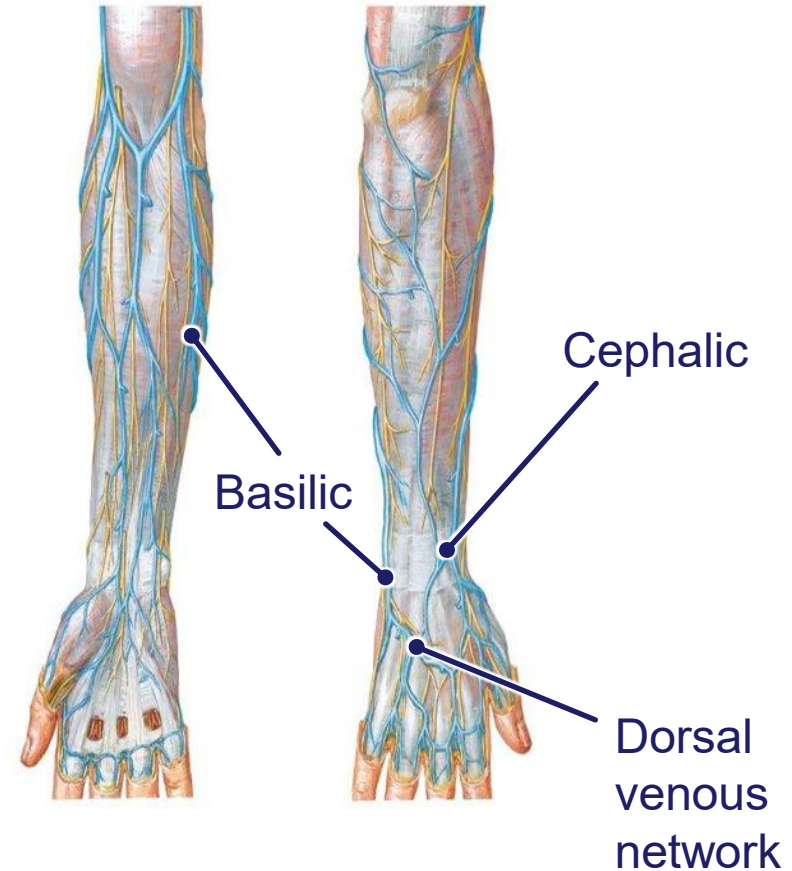
Superficial and Deep venous drainage of the hand

• Superficial veins

- On the dorsum of the hand, the **dorsal venous network** coalesces on the **medial side** of the wrist to form the **basilic vein** and on the **lateral side** of the wrist to form the **cephalic vein**

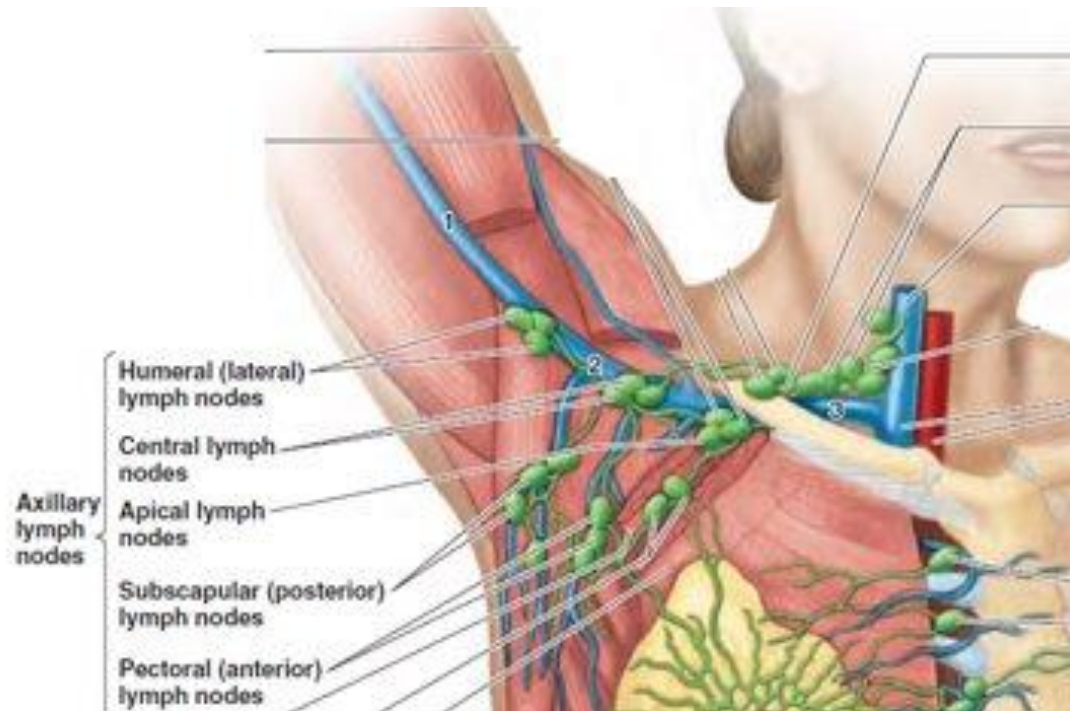
• Deep veins

- Superficial and deep **venous palmar arches** accompany **arterial palmar arches** in the hand
- Drain into the **paired radial and ulnar veins**



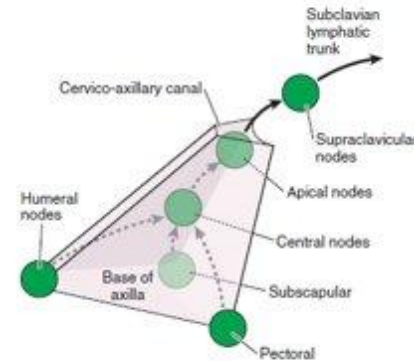
Lymphatic Drainage of the Upper Limb

- Like venous drainage, lymphatic drainage can be divided into **deep and superficial**
- **Follows the same path of the venous drainage of the upper limb**
- **All** lymphatics will drain into the axillary nodes

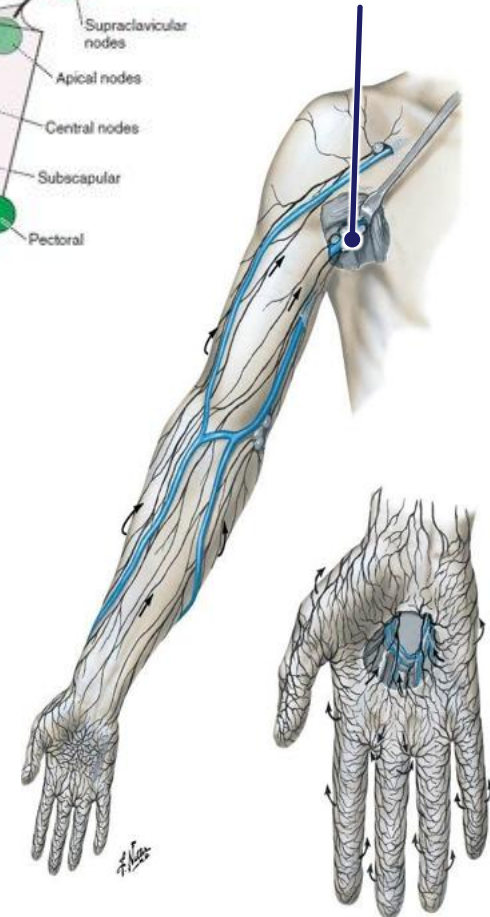


Superficial Lymphatic Drainage of the Upper Limb

- Superficial drainage: veins within the skin and superficial fascia
 - **Medial** hand, forearm, arm:
 - Lymphatic fluid will go to **lateral (humeral) axillary nodes**
 - then to **central axillary nodes**
 - then finally to **apical axillary nodes**
 - **Lateral** hand, forearm, arm:
 - Lymphatic fluid will go **directly** to the **apical axillary nodes**
- All lymph then pass from **apical** to **subclavian trunk** to **lymphatic/thoracic duct**

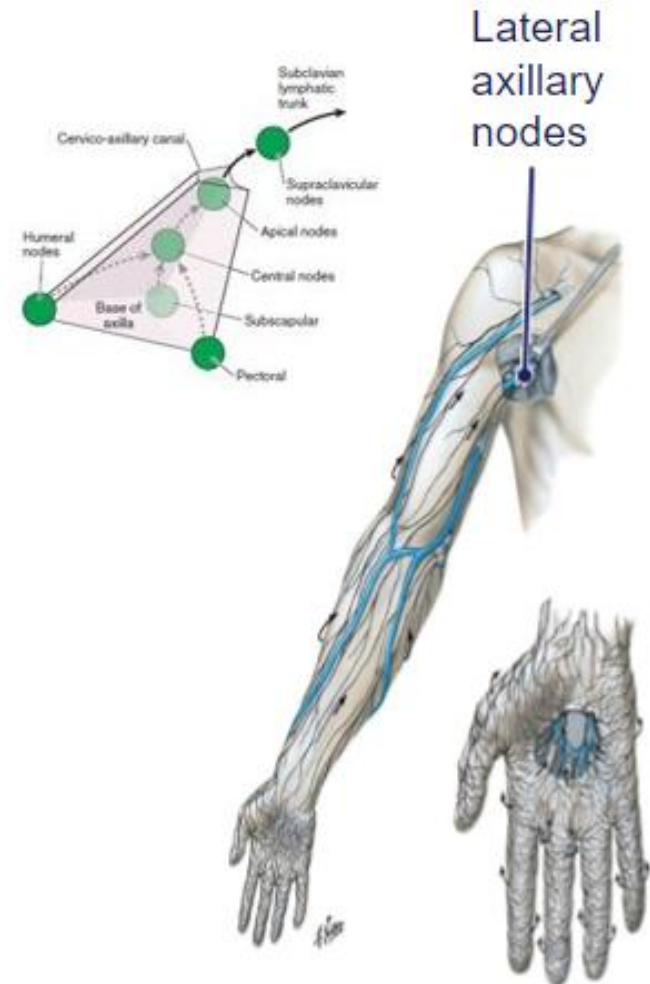


Lateral axillary nodes

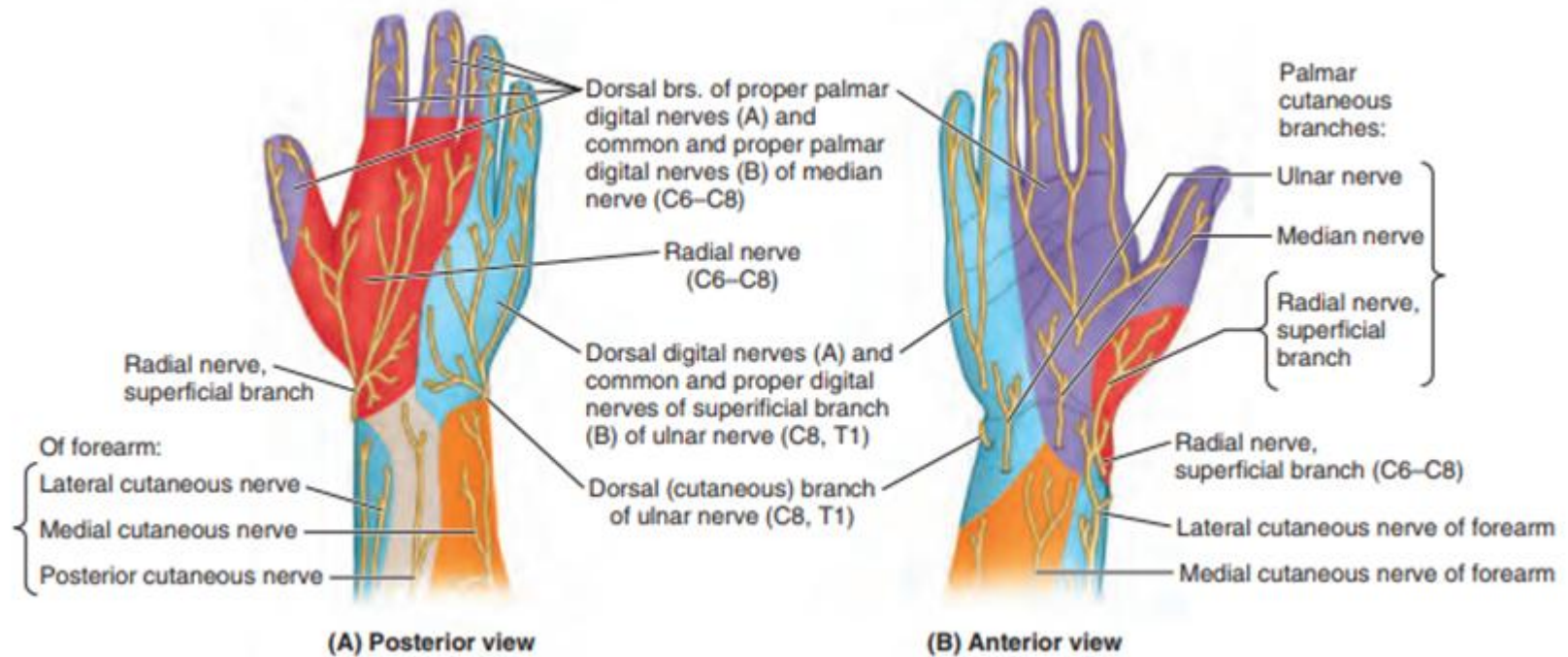


Deep Lymphatic Drainage of the Upper Limb

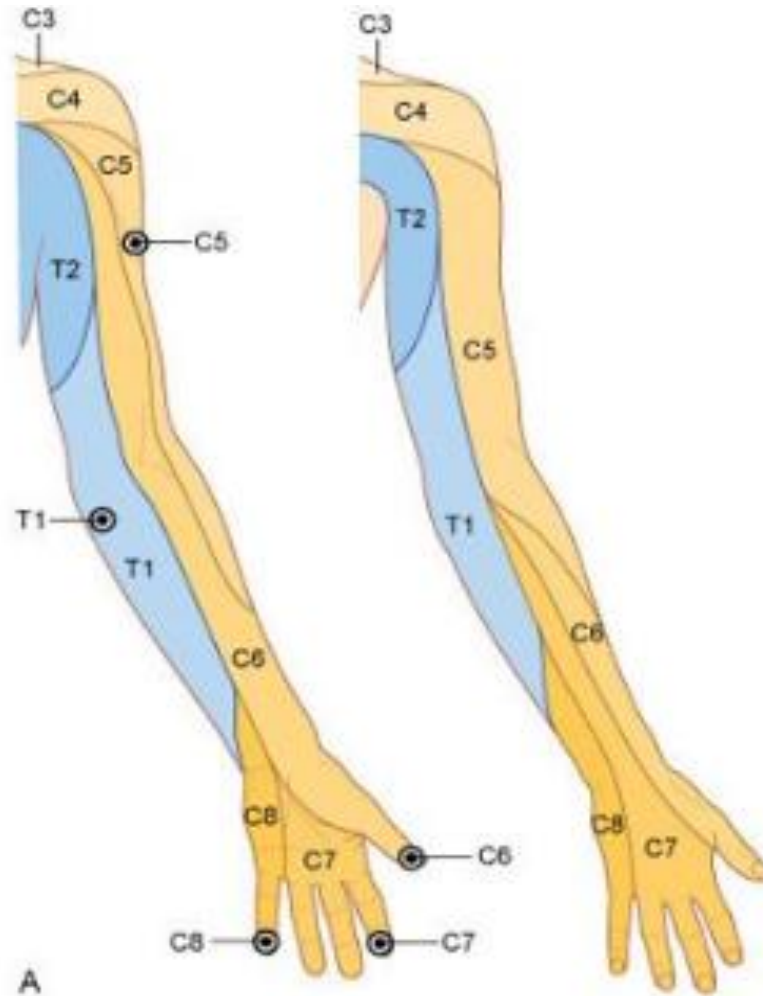
- Deep drainage: veins within the compartments
- Compartments of hand, forearm, and arm:
 - Lymphatic fluid will go to **lateral (humeral) nodes**
 - then to **central axillary nodes**
 - then finally to **apical axillary nodes**
- All lymph then pass from **apical** to **subclavian trunk** to **lymphatic/thoracic duct**



Cutaneous Innervation of the Hand



Dermatomes of the Upper Limb



Please note that there are several dermatome maps. Please use the one in this lecture.

Complete the table (for self-review)

Movement of the thumb	Nerves innervating	Muscles acting
Flexion		
Extension		
Abduction		
Adduction		
Opposition		

Complete the table (for self-review)

Movement of the MCP joint	Nerves innervating	Muscles acting
Flexion		
Extension		
Abduction		
Adduction		

Complete the table (for self-review)

Movement of the PIP joints of digits 2-5	Nerves innervating	Muscles acting
Flexion		
Extension		

Movement of the DIP joints of digits 2-5	Nerves innervating	Muscles acting
Flexion		
Extension		

fin